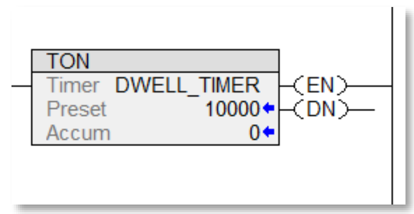
TIMER INSTRUCTIONS SUMMERY

RSLogix 500 TIMERS

Timer On Delay (TON)

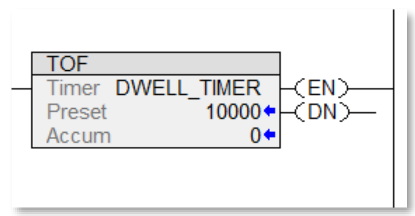


The Timer On Delay (**TON**) is an output instruction that will turn **on** a controlled output when the accumulated value is equal to the preset value.

The equal values will result in the **Done** bit being **set**. The Done bit may be used to control other logic.

While the rung is true, the timer will begin to elapse at time base intervals at 0.001 of a second.

Timer Off Delay (TOF)

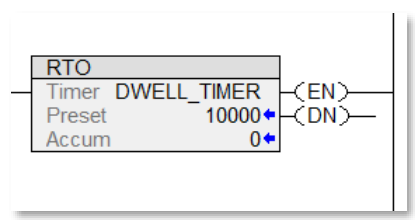


The Timer Off Delay (**TOF**) is an output instruction that will turn **off** a controlled output when the accumulated value is equal to the preset value.

The equal values will result in the **Done** bit being **reset**. The Done bit may be used to control other logic.

While the rung is false, the timer will begin to elapse at time base intervals at 0.001 of a second.

Retentive Timer On (RTO)



A Retentive Timer On (**RTO**) is similar to the Timer On Delay. The difference is the Retentive Timer will *retain its accumulated* value when the rung is false. When the rung is true, the accumulated value will increment from the current value.

When the accumulated and preset value are equal, the **Done** bit is **set**.

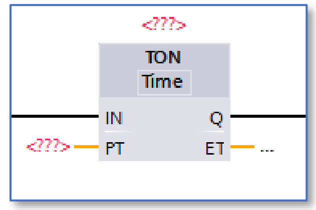
STATUS BITS

Enable Bit **.EN**	**TRUE** when timer rung True
Timer Timing **.TT**	**TRUE** while timer is elapsing
Timer On (TON) and Retentive Timer On (RTO)	
Done Bit **.DN**	**TRUE** when Preset and Accumulated are equal
Timer Off (TOF)	
Done Bit **.DN**	**FALSE** when Preset and Accumulated are equal

Siemens TIMERS

Timer On Delay (TON)

The Timer On Delay (**TON**) is an instruction that will turn **on** a controlled output when the Elapsed Time is equal to the Preset Time.

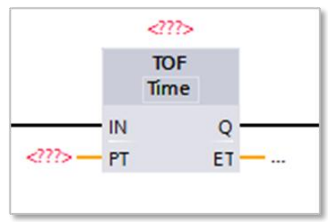


The equal values will result in the Output Bit being **set**. The Output Bit is used to control other logic.

While the rung is true, the timer will begin to elapse at intervals of 0.001 seconds, milliseconds.

Timer Off Delay (TOF)

The Timer Off Delay (**TOF**) is an output instruction that will turn **off** a controlled output when the elapsed time is equal to the preset time.

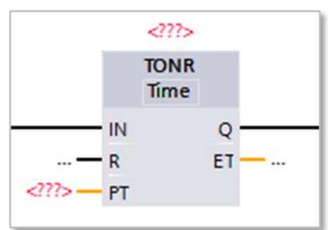


The equal values will result in the Output Bit being **reset**. The Output Bit is used to control other logic.

While the rung is false, the timer will begin to elapse at intervals of 0.001 seconds, milliseconds.

Timer On Retentive (TONR)

A Timer On Retentive (**TONR**) is similar to the Timer On Delay. The difference is the Timer On Retentive will *retain its Elapsed Time* when the rung is false. When the rung is true, the Elapsed Time will increment from the current value.



When the Elapsed Time and Preset Time are equal, the Output Bit is **set**.

STAUS BITS

In Bit **.IN**	**TRUE** when timer network True
Timer On (TON) and Retentive On Timer (TONR)	
Output Bit **.Q**	**TRUE** when Preset and Accumulated are equal
Timer Off (TOF)	
Output Bit **.Q**	**FALSE** when Preset and Accumulated are equal

RUNG OR NETWORK STATE FOR ELASPING TIMER

For a Timer On Delay (**TON**) the initial condition for the rung is **FALSE** and transitions to **TRUE**.

For a Timer Off Delay (**TOF**) the initial condition for the rung is **TRUE** and transitions to **FALSE**.

The final state of the rung determines if a **TON** or **TOF** timer is used.

> If the rung or network state is **TRUE**, then to have the timer elapse a **TON** is used.
>
> If the rung or network state is **FALSE**, then to have the timer elapse a **TOF** is used.

TIMER ON and TIMER OFF DELAY and DELAY OFF EXAMPLES

These circuits use generic rung/network examples

TIMER ON DELAY ON (Energize Timer Delay On)

```
 INPUT
----]  [--------------------------------------------------------------------------(TIMER ON)-

 TIMER ON.STATUS BIT                                                              OUTPUT
----]  [-----------------------------------------------------------------------------(   )---
```

TIMER ON DELAY OFF (Energize Timer Delay Off)

```
 INPUT
----]  [--------------------------------------------------------------------------(TIMER ON)-

 TIMER ON.STATUS BIT                                                              OUTPUT
----] / [----------------------------------------------------------------------------(   )---
```

TIMER OFF DELAY OFF (Deenergize Timer Delay Off)

```
 INPUT
----]  [------------------------------------------------------------------------(TIMER OFF)-

 TIMER OFF.STATUS BIT                                                             OUTPUT
----]  [-----------------------------------------------------------------------------(   )---
```

TIMER OFF DELAY ON (Deenergize Timer Delay On)

```
  INPUT
----]  [--------------------------------------------------------------------------------------(TIMER OFF)-

 TIMER OFF.STATUS BIT                                                                                  OUTPUT
----⌋ / ⌊--------------------------------------------------------------------------------------------------(   )---
```